

Activity 2: SQL JOINS Challenge

1. Display movies with their director names. Show movie title and director name.

```
SELECT name, title FROM movies NATURAL JOIN directors;
```

2. Find actors who appeared in movies directed by Christopher Nolan. Show actor name and movie title.

```
SELECT a.actor_name, m.title  
FROM actors a  
JOIN movies m ON a.movie_id = m.movie_id  
JOIN directors d ON m.name = d.name  
WHERE d.name = 'Christopher Nolan';
```

3. Display movies with their studio names and total revenue (domestic + international). Show movie title, studio name, and total revenue.

```
SELECT m.title, s.studio_name, (b.domestic_revenue + b.international_revenue)  
AS total_revenue  
FROM movies m  
JOIN studios s  
ON b.studio_id = s.studio_id  
JOIN box_office b  
ON m.movie_id = b.movie_id;
```

4. Find movies where Leonardo DiCaprio was the lead actor (billing_order = 1). Show movie title and his role.

```
SELECT  
m.title,  
FROM movies m  
JOIN movie_cast c  
ON m.movie_id = c.movie_id  
JOIN actors a  
ON c.actor_id = a.actor_id  
WHERE a.name = 'Leonardo DiCaprio'  
AND c.billing_order = 1;
```

5. Count how many movies each director has made. Show director name and movie count. Include directors who haven't made any movies yet. Order by movie count (highest first).

```
SELECT  
d.name AS director_name,  
COUNT(m.movie_id) AS movie_count  
FROM directors d  
LEFT JOIN movies m  
ON d.director_id = m.director_id  
GROUP BY d.director_id, d.name  
ORDER BY movie_count DESC;
```

6. Find actors who have appeared in more than one movie. Show actor name and number of movies they've been in.

```
SELECT  
a.name AS actor_name,  
COUNT(DISTINCT c.movie_id) AS movie_count  
FROM actors a  
JOIN cast c  
ON a.actor_id = c.actor_id  
GROUP BY a.actor_id, a.name  
HAVING COUNT(DISTINCT c.movie_id) > 1  
ORDER BY movie_count DESC;
```

7. Display movies released after 2010 with their directors and studios. Show movie title, release year, director name, and studio name.

```
SELECT  
m.title,  
m.release_year,  
d.name AS director_name,
```

```
s.studio_name
FROM movies m
JOIN directors d
ON m.director_id = d.director_id
JOIN studios s
ON m.studio_id = s.studio_id
WHERE m.release_year > 2010;
```